

they use SiGe technology.

Infineon's SiGe based Radar System IC (RASIC) series consists of a group of highly-integrated functions for 76GHz to 77GHz range for automotive radar.

GPS, V2X and everything else.

While these are more for speaking than merely listening, GPS and V2X enable an intelligent transport system where all vehicles and infrastructure systems are interconnected with each other. This connectivity aims to provide

The standardised frequency band of 5.9GHz (IEEE 802.11p) derived from Wi-Fi as the technology base for wireless communication enables direct, low-latency information exchange between vehicles and infrastructure. IEEE 1609 family of standards for Wireless Access in Vehicular Environments (WAVE) defines the architecture, communications model, management structure, security mechanisms and physical access for high-speed (up to 27Mb/s),

match to ensure a precise positioning accuracy, low latency and dead reckoning.

U-blox lists an automotive-grade V2X transceiver module called THEO-P1 series for infrastructure and vehicles, which is in compliance with WAVE and ETSI ITS G5 for American and European operations. U-blox claims that this chip has a range of one kilometre.

Sense of awareness

While having cameras to see details on the road are important, those that come with processing software to pick up lane markers and monitor your distance from these are what enable cars and other vehicles to drive themselves. If you stray over the line without signalling, you will hear a warning tone or feel a physical alert such as a vibration in the steering wheel or seat. More advanced lane-keeping assist systems selectively apply brakes or nudge the steering to guide you back if you are wandering.

In autonomous mode, such as in Tesla with autopilot, a front-facing radar, a camera with image-recognition capability and 360-degree ultrasonic sonar will pick up lane markings on the road as well as other vehicles to steer and control vehicle speed.

In a talk delivered at Electronics Rocks 2016 conference, Dr S. K. Sinha said, "Thirty years ago, software in the cars was zero—there was no software. And, in just a short period of 30 years, the amount of software has gone to ten million lines of code, in low-end cars. If you talk about Audi, Mercedes or BMW cars, these have hundred million lines of code today. Thousands of functions are being realised with software today."

"The biggest worry is if the software controlling the cars have enough information about

Driverless cars may be safe, but are these secure

Although this means that car security is at risk all the time, automobile manufacturers have also taken some steps to safeguard vehicles. One such initiative is known as E-Safety Vehicle Intrusion Protection Applications (EVITA), which was developed by car manufacturers in the European Union. "This is a hardware solution. They have made hardware that has encryption facilities and all that, and which can be included into any microcontroller intended for automotive applications," he said. There is also a software solution called OVERSEE (an open vehicular secure platform), which involves the separation of software applications from hardware components.

—Dr S.K. Sinha in a talk delivered at Electronics Rocks 2016

Up next, trucks

The race is already on to put driverless trucks on public roads. In May 2015, the first self-driving truck rolled through Nevada. Of the several tests around the world since then, the most interesting is a convoy that drove across Europe using a new technology called platooning. It connects trucks using Wi-Fi, sensors, GPS and cameras and allows the leading vehicle to dictate speed and direction, while the rest automatically steer, speed up and slow down.

more precise knowledge of the traffic situation across the entire road network.

Continental has been at it since 2006. Their system can be complemented by the intelligent localisation sensor M2XPro or Motion Information to X Provider. Fusion of sensor data about speed, steering angle, GNSS and inertial information, along with position and timing information are essential for V2X communication to detect and evaluate potential threats. This enables a perception, detection and assessment of road obstacles and dangerous situations even before these can be noticed optically.

short-range (up to 1000m), low-latency wireless communications in the vehicular environment.

Primary architectural components defined by these standards are OnBoard Unit (OBU), Road Side Unit (RSU) and WAVE interface. Dedicated Short-Range Communication (DSRC) is a communication technology that works in the 5.9GHz band in the USA and China or 5.8GHz band in Japan and Europe.

ST has a V2X system using their co-developed Autotalks V2X Communication processor along with ST's multi-constellation GNSS receiver, Teseo, that builds a perfect